

Boardwise — an AI agent that writes and debugs firmware

The Problem

Embedded development has not changed in twenty years. A firmware engineer writes C against a vendor SDK, flashes it to a board, watches a serial log, and guesses. A single peripheral misconfiguration can cost two days.

The tooling that transformed web and backend development has skipped this field entirely, because firmware cannot be validated without hardware. Code that compiles tells you almost nothing about whether it will run on the chip in front of you.

Our Solution

Boardwise writes, flashes, tests and debugs firmware against real and simulated hardware.

The agent generates code from a plain-language requirement, builds it for the target chip, runs it on a physical board in our hardware farm or in a cycle-accurate simulator, reads the trace output and register state, and iterates until the behaviour matches the requirement.

It supports 23 microcontroller and system-on-chip families. When a peripheral misbehaves, it inspects register state directly rather than inferring from logs — which is how an experienced engineer debugs, and what every code assistant built for software cannot do.

The Market

There are roughly 1.4 million embedded developers worldwide. Development tooling for them is a \$2.1B market, small next to general software tooling but underserved and growing with device volume.

Our entry segment is industrial and automotive suppliers, where firmware teams are small, deadlines are fixed, and a week saved on bring-up is worth more than the tool costs in a year.

Competition

General-purpose coding assistants produce plausible firmware that fails on hardware, because they have no feedback loop to the device. Vendor toolchains are excellent at compiling and poor at everything else.

The barrier here is not the model. It is the hardware farm, the per-chip instrumentation, and the debugging harness — eighteen months of work that does not look like AI and cannot be skipped.

The Team

Rei Tanaka spent nine years in embedded engineering, including four as firmware lead on a consumer device family that shipped over 40 million units. Sofia Berg built compiler and toolchain infrastructure at a semiconductor vendor and maintained the debug probe stack used across its product line.

They built the first version of the hardware farm in Rei's garage over a year before incorporating. Eleven people today, nine of them engineers.

Traction

- 2,100 developers have run at least one build
- 340 use it weekly
- median bring-up time for a new peripheral fell from 6.5 hours to 40 minutes across instrumented sessions
- 23 chip families supported, up from 4 at launch
- three industrial suppliers running paid pilots

Business Model

Seat-based subscription for individual developers and an enterprise tier that includes private hardware targets and on-premise deployment. Pricing is published at \$90 per seat per month.

Enterprise pilots are converting at a rate we consider promising, though our sample is still three.

Go-To-Market

Bottom-up adoption through developer communities and conference workshops, then expansion into the employing organisation. The three pilots all began with an individual engineer who brought us to their team.

Funding

We are raising \$9M to double the hardware farm, add automotive-grade chip families, and build the enterprise deployment path that the pilots are asking for.